The tool is just the hammer. The platform is the house.

Build with AI speed. Govern with enterprise confidence.

CHAPTER 01 · MATURITY MODEL

The Innovative Maker is the game changer.

Business Innovators start in **vibe.powerapps.com** — describe an idea, ship a working app. Pro Developers ship in **VS Code + GitHub Copilot** with full CI/CD. In between sits the persona that breaks the old ceiling: the **Innovative Maker**, who fluidly moves between vibe.powerapps.com and VS Code — *given incredible AI superpowers, but contained and controlled by the platform.*

Business Innovators ideate · Innovative Makers cross over · Pro Developers harden · Power Platform governs every stage.

1 IDEATE & BUILD

Business Innovator

vibe.powerapps.com

- ✓ Describe the idea in natural language
- ✓ AI builds the first working app
- ✓ Validate with real users in days
- ✓ Hand off — or keep going

★ GAME CHANGER

2 CROSS-OVER · CONTAINED SUPERPOWERS

Innovative Maker

vibe.powerapps.com **and** VS Code + GitHub Copilot

- ✓ Starts in vibe.powerapps.com *or* VS Code — whichever fits the task
- ✓ Pulls Code Apps into VS Code when vibe hits a ceiling — and keeps building from there
- ✓ Uses GitHub Copilot Agent Mode to extend, refactor, and harden at AI speed
- ✓ Guardrails via `copilot-instructions.md` and `AGENTS.md`
- ✓ **Same artifact, same Dataverse, same governance — either way**

3 HARDEN & SCALE

Pro Developer

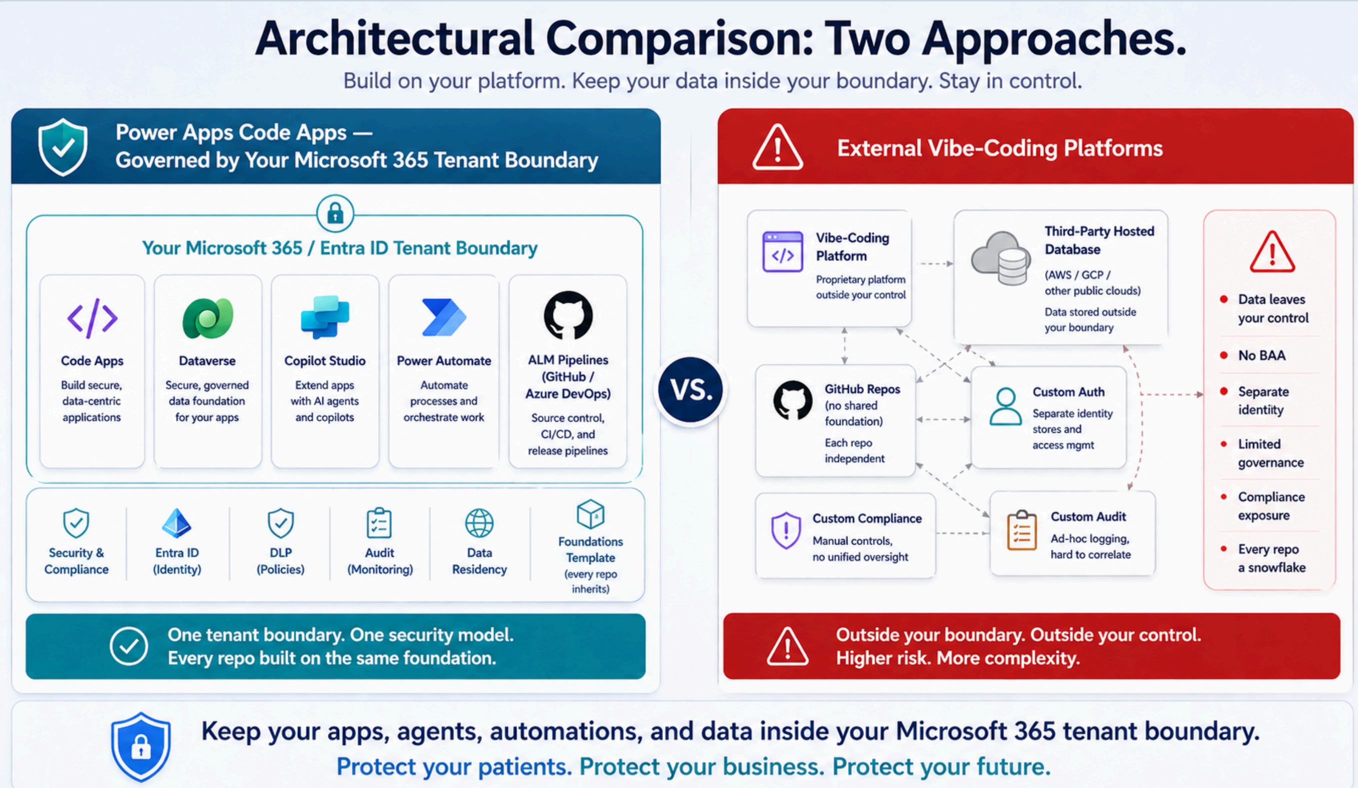
VS Code + GitHub + ALM

- ✓ Custom components, integrations, complex logic
- ✓ Source control, code review, CI/CD pipelines
- ✓ Performance, security, observability
- ✓ Ship to Managed Environments with confidence

These are roles, not necessarily different people. Often a Business Innovator *becomes* the Innovative Maker the moment vibe hits its limit — and the platform makes that step continuous instead of a cliff. The Pro Developer joins when CI/CD, custom components, or hardening matter. Same solution, same Dataverse, same governance throughout — **that's what makes the cross-over safe.**

Configure once. Let 1,000 makers self-serve.

Admins configure ALM pipelines and **Managed Environments** once. After that, makers ship to dev → test → prod with proper promotion, DLP, and approval gates baked in.



Architectural comparison — build on your platform, keep your data inside your tenant boundary, stay in control.

✓ Code Apps + Managed Environments

- ✓ One-time admin setup of pipelines, DLP, and environment groups
- ✓ Makers self-serve deployments — guardrails enforced automatically
- ✓ **Every repo inherits the same governed foundation** — instructions, ALM, security, tests, agent guardrails — from a versioned template (see [Chapter 10](#))
- ✓ Update the foundation once → every downstream repo can pull it in
- ✓ Solution packaging carries data model + UI + logic together
- ✓ Tenant-wide audit, telemetry, and CoE visibility across the fleet

External Vibe Platforms — Repos Without a Foundation

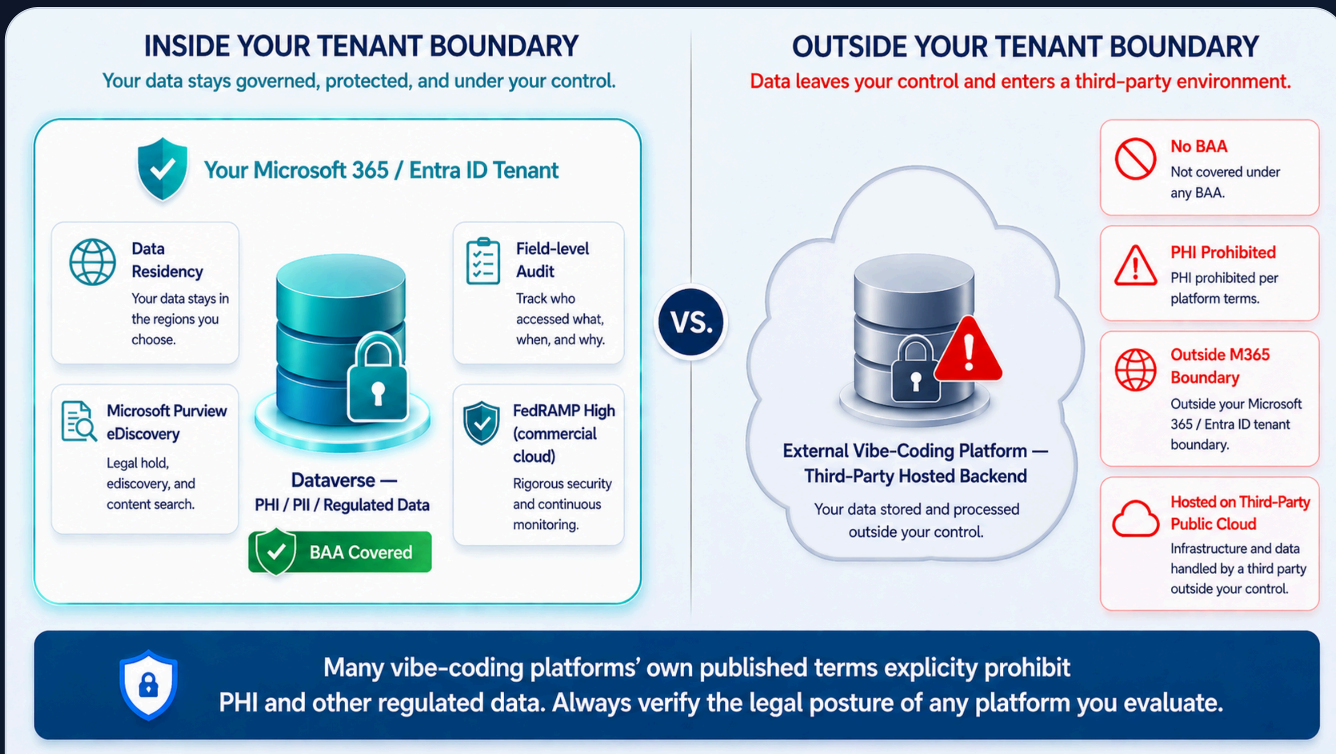
- ✕ **Every repo starts from scratch** — no inherited instructions, no inherited ALM, no inherited tests
- ✕ Governance must be re-invented per repo, per team
- ✕ Each external data backend (hosted Postgres, Firebase, Supabase, etc.) is its own snowflake
- ✕ No tenant-wide DLP, no native solution packaging, no way to push a fix across the fleet
- ✕ **Estimated +1.5 to +2.5 FTEs** of platform/governance overhead at scale (illustrative, varies by org)

The problem isn't repo count — Code Apps create plenty of repos too. The problem is repos that don't inherit a common foundation.

⚠ CHAPTER 03 · THE BOUNDARY

This is not a feature comparison. It is a boundary.

For any organization handling PHI, this single chapter ends the conversation. It's a contractual and regulatory boundary — not a preference.



Data must stay inside your tenant boundary — protect patient data, preserve compliance, maintain trust.

✓ Inside the Fence

CODE APPS WITH DATAVERSE

- ✓ Microsoft-operated SaaS governed by your **Microsoft 365 / Entra ID tenant boundary**, with regional data residency
- ✓ Covered by a **signed Microsoft BAA**
- ✓ PHI, PII, and regulated workloads explicitly supported
- ✓ FedRAMP High and HITRUST attestations apply on the Microsoft commercial cloud; sovereign-cloud options (GCC, GCC High, DoD) available — verify Code Apps availability in your target cloud
- ✓ Field-level audit and change tracking native to Dataverse; integrated with Microsoft Purview eDiscovery

Outside the Boundary

EXTERNAL VIBE-CODING PLATFORMS

- ✕ Default data layer is a **third-party hosted backend** (hosted Postgres, Firebase, Supabase, etc.) on a public cloud you don't control — outside your Microsoft 365 tenant boundary
- ✕ No Microsoft BAA coverage
- ✕ Many of these platforms' **own published terms explicitly prohibit PHI** — see proof drawer below for one transparent example
- ✕ Cannot be easily relocated into a customer-controlled Microsoft cloud boundary

This pattern is common across the category. We use **Lovable** here as a representative example because their terms are particularly explicit. Replit, Bolt, v0, and similar platforms typically carry comparable restrictions in their own published terms — always verify the current legal documents of any platform you evaluate.

TERMS & CONDITIONS lovable.dev/terms (https://lovable.dev/terms)

*"You agree not to upload, input, or otherwise provide any **protected health information under HIPAA**, or any other sensitive categories of data... Our Services are **not designed to handle that type of data**, and we disclaim all responsibility if you choose to submit it."*

DATA PROCESSING AGREEMENT · §8

lovable.dev/data-processing-agreement (https://lovable.dev/data-processing-agreement)

*"The Customer shall **not provide any data to Lovable which is classified as sensitive**. For the avoidance of doubt, the Customer agrees not to upload... any **protected health information under HIPAA**..."*

PRIVACY POLICY lovable.dev/privacy (https://lovable.dev/privacy)

Explicitly warns against uploading sensitive health data.

KEY TAKEAWAY FOR HEALTHCARE & LIFE SCIENCES

Microsoft signs a BAA and keeps your data inside your Microsoft 365 tenant boundary. The **external vibe-coding category, broadly, does not** — and many of these platforms explicitly prohibit PHI in their own published terms. Always confirm the legal posture of any platform before letting regulated data near it.

CHAPTER 04 · AGENTIC AI

One intelligent connection for agents.

In the era of intelligent agents, the data layer must speak the agent's language. Many databases now ship MCP servers — including Azure SQL, Cosmos DB, and Postgres — but they expose *schema*. **Dataverse's MCP exposes business semantics:** security roles, business units, business rules, choice sets, relationships, and audit — the things agents actually need to act safely on enterprise data.

DATAVERSE

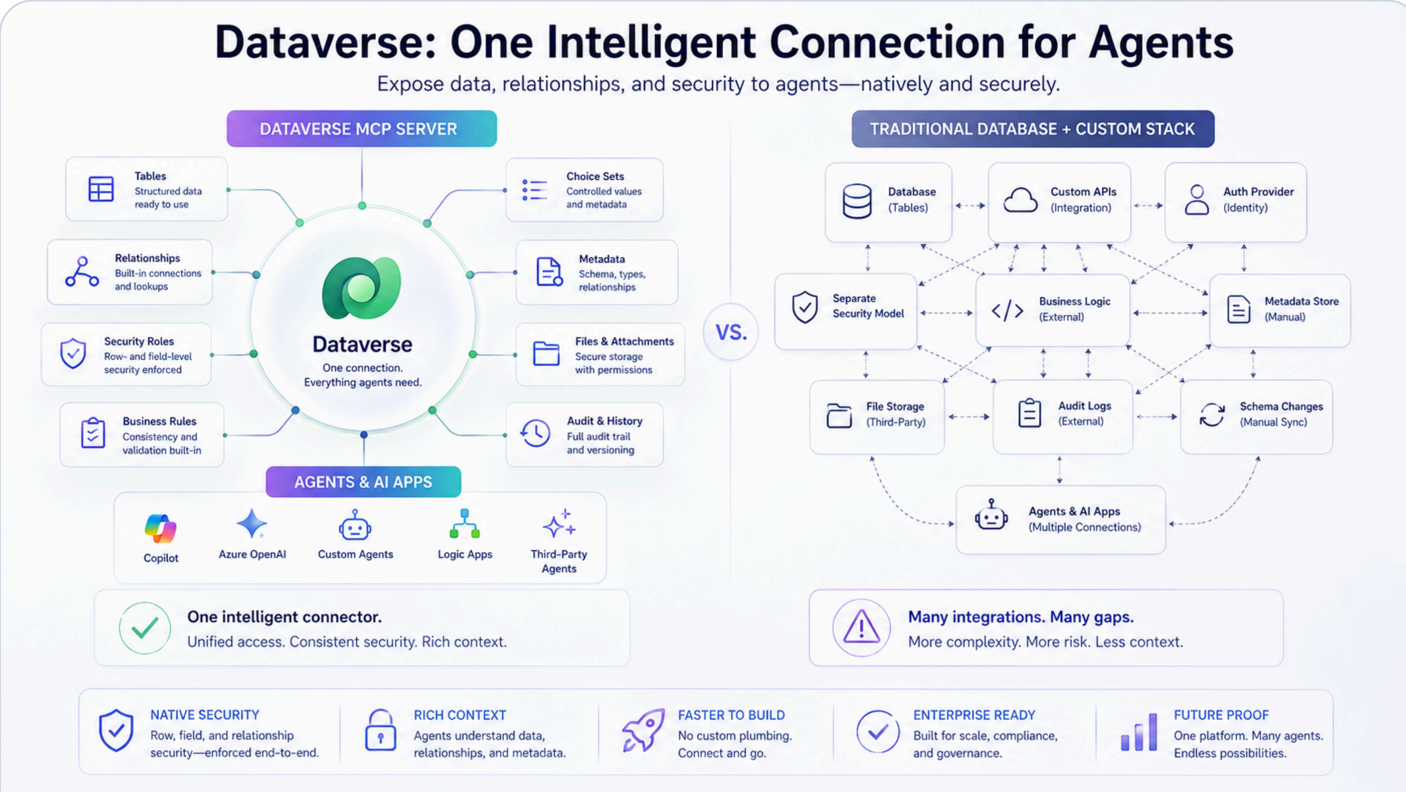
Total context. One connection.

- ✓ Auto-discover tables, columns, relationships
- ✓ Row & column security flow through
- ✓ Business rules & choice sets introspectable
- ✓ Copilot Studio, M365 Copilot, custom agents

SCHEMA-ONLY MCP

Plumbing without meaning.

- ✕ MCP exposes columns and types, not business rules or security roles
- ✕ Row/column security must be re-described to every agent
- ✕ Constant drift between schema, prompts, and policy
- ✕ Each new agent restarts the integration cost

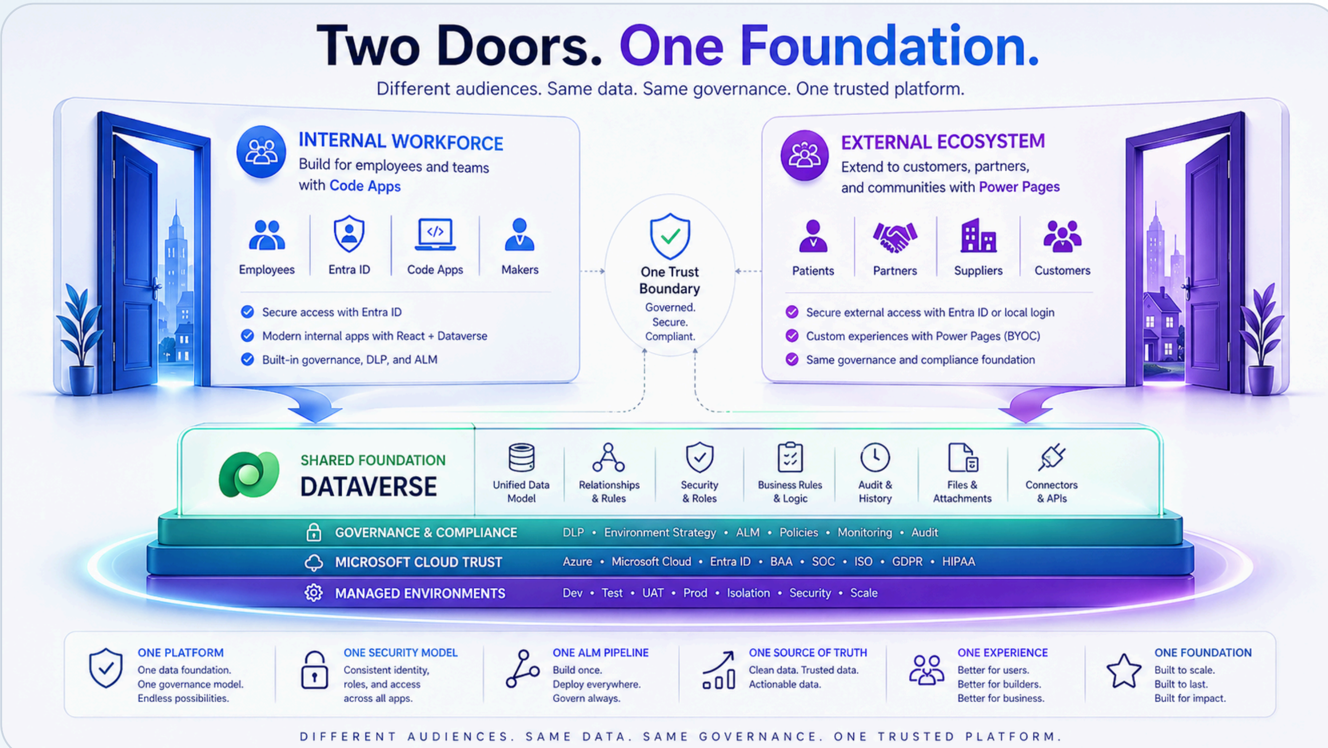


Dataverse exposes tables, relationships, metadata, security, and business rules — natively to agents.

CHAPTER 05 · INTERNAL & EXTERNAL

Two Doors. One Foundation.

Different audiences. Same data. Same governance. One trusted platform.



⌂ Internal Workforce

CODE APPS · ENTRA ID

- ✓ Modern internal apps with React + Dataverse
- ✓ Secure access with Entra ID
- ✓ Built-in governance, DLP, and ALM

External Ecosystem

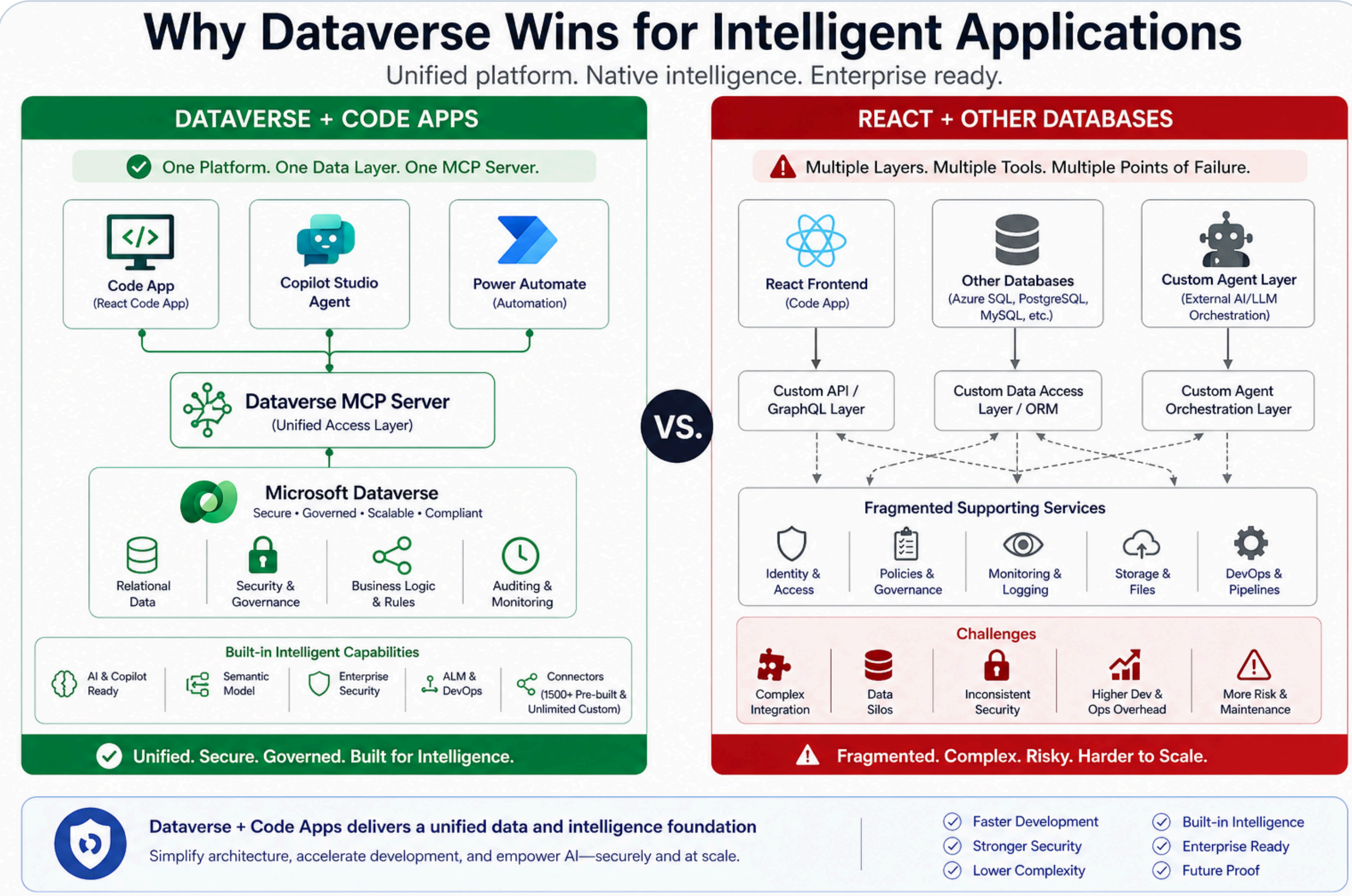
POWER PAGES · BYOC

- ✓ Customers, patients, partners, suppliers
- ✓ Modern React on the secure Power Pages runtime
- ✓ Same Dataverse security, end-to-end

ONE PLATFORM One data foundation	**ONE SECURITY** Identity, roles, access
ONE ALM Build once. Govern always.	**ONE TRUTH** Clean, trusted data
ONE EXPERIENCE Users · Builders · Business	**ONE FOUNDATION** Built to scale & last

A traditional database is just a database. Dataverse is a platform.

Dataverse is a governed business application platform. The difference shows up in nearly every dimension that matters at enterprise scale.



Why Dataverse wins for intelligent applications — unified platform, native intelligence, enterprise ready.

CAPABILITY	DATAVERSE	OTHER DATABASES
Row & column security	✓ Built-in security roles, business units, hierarchical, field-level	✕ Custom-built with views, RLS policies, and app-tier code
MCP Server	✓ Native — exposes business semantics: security roles, business	✕ Where MCP exists, exposes *schema only* — security and

	rules, choice sets, relationships, audit	business rules must be re-described per agent
Agent intelligence	✓ Copilot Studio + M365 Copilot speak Dataverse natively	✕ Requires bespoke prompt engineering and middleware
Business logic location	✓ Centralized: business rules, plug-ins, flows	✕ Scattered across app code, stored procs, triggers, services
ALM & solution packaging	✓ First-class solutions carry schema + UI + logic together	✕ DACPAC + custom scripts + bespoke release pipelines

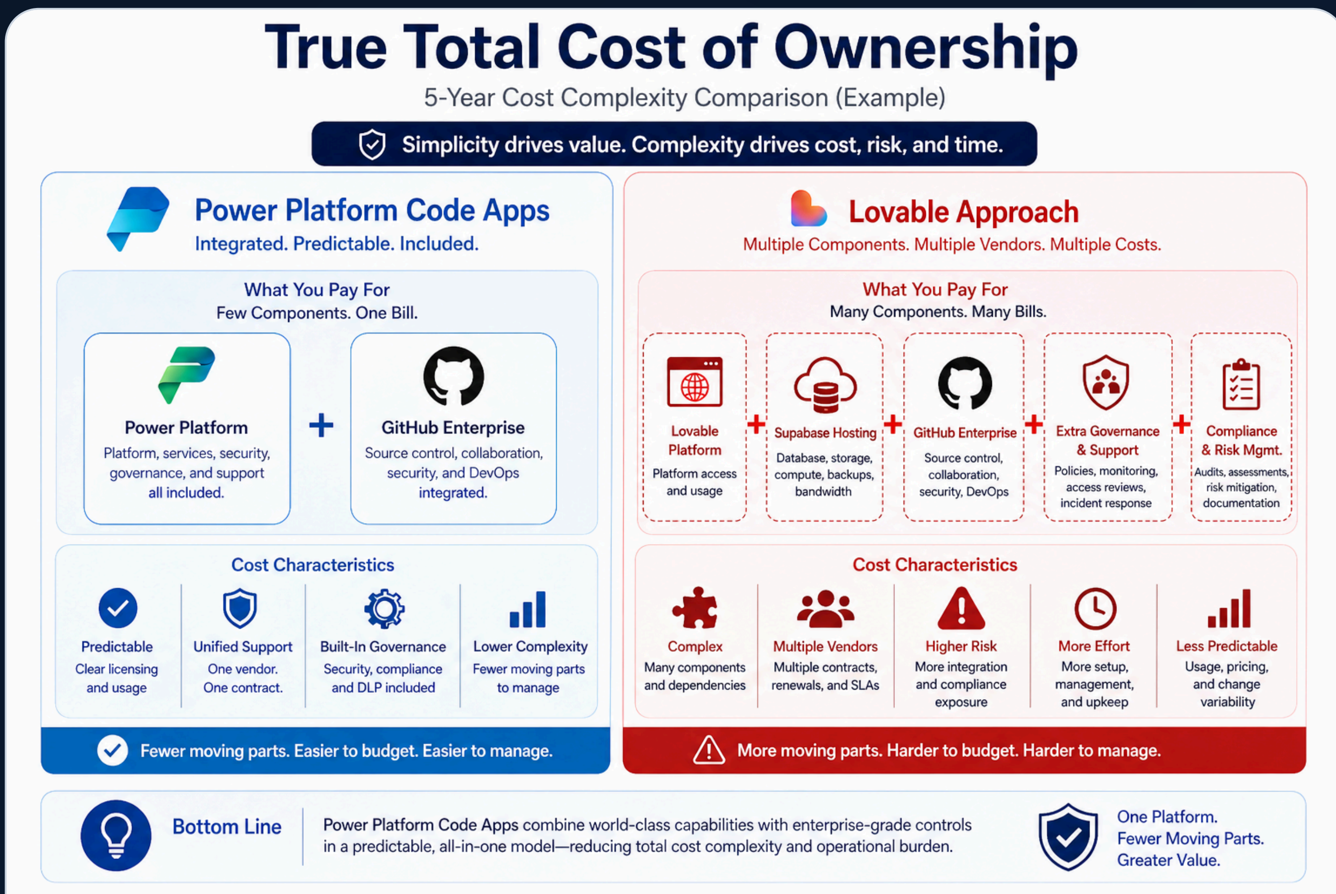
Show 4 more rows ▾

CHAPTER 07 · TRUE TOTAL COST OF OWNERSHIP

The hidden cost of tool sprawl.

Simplicity drives value. Complexity drives cost, risk, and time.

We deliberately don't put dollar signs here — pricing varies by scale, region, and deal. The point is the **shape** of the cost structure.



True total cost of ownership — simplicity drives value, complexity drives cost, risk, and time.

POWER PLATFORM CODE APPS

Integrated. Predictable. Included.

What you pay for: Few components. One bill.

PP **Power Platform**
Platform, services, security, governance, support — all included

GH **GitHub Enterprise**
Source control, collaboration, security, DevOps

Predictable
Clear licensing

Unified Support
One vendor

Built-in Governance
Compliance & ALM

Future-Compatible
Fewer parts

EXTERNAL VIBE PLATFORM APPROACH

Many components. Many bills.

What you pay for: Many components. Many bills.

- **VP** — **Vibe-Coding Platform**
  Subscription, usage, compute, backups
- **DB** — **Third-Party Data Backend**
  Hosted database on AWS / GCP / other public clouds — storage, networking
- **GH** — **GitHub Enterprise**
  Repos, security, governance, DevOps
- **+G** — **Extra Governance & Support**
  Policies, monitoring, incident response, added FTEs
- **C&R** — **Compliance & Risk Mgmt**
  Audit, logging, certifications, documentation

Complex Multiple bills	**Higher Risk** More integration surface
Reduced Reliability Less consistency	**Higher Maintenance** Vendor sprawl

BOTTOM LINE

One platform. Fewer moving parts. Greater value.

CHAPTER 08 · AI FLEXIBILITY

Bring any frontier model. Always know which one you're using.

Coding-model leadership shifts month to month. The **VS Code ecosystem** — GitHub Copilot for first-party Microsoft-supported models, plus Cursor, Cline, and other agent tools for the broader frontier — lets developers pick the model by name and version, and swap the moment a stronger one ships. With most external vibe-coding platforms, the model is opaque.

Below: a cross-section of leading coding models available across the VS Code ecosystem in 2026. Availability inside GitHub Copilot itself varies by model and changes over time — check the current GitHub Copilot documentation (https://docs.github.com/en/copilot) for the live list.

ANTHROPIC

Claude Opus 4.x

Sonnet 4.x · Haiku 4.5

Industry-leading agentic coding & long-horizon refactors.

OPENAI

GPT-5 / GPT-5 Codex

o-series reasoning

Strong general coding & deep reasoning workflows.

GOOGLE

Gemini 2.5 Pro

Gemini Code Assist

Massive context windows for whole-repo reasoning.

GITHUB

GitHub Copilot

Multi-model: Claude · GPT · Gemini

One subscription, you choose the brain per task.

CURSOR

Cursor + Composer

Routes to Claude / GPT / Gemini

IDE-native agent loops, model of your choice.

XAI

Grok Code

Grok 4 family

Real-time web context for current API knowledge.

DEEPSEEK

DeepSeek-Coder V3

Open-weights option

Strong code completion at lower cost-per-token.

ALIBABA

Qwen Coder

Open-weights, multilingual

Self-hostable for sovereign workloads.

The cadence is the point: new frontier coding models drop every few weeks. Within **days to weeks** of release, they typically appear in at least one VS Code-compatible surface — GitHub Copilot, Cursor, Cline, or direct API integrations. Yesterday's leader rarely stays the leader for long — and your developers will always know which one they're running.

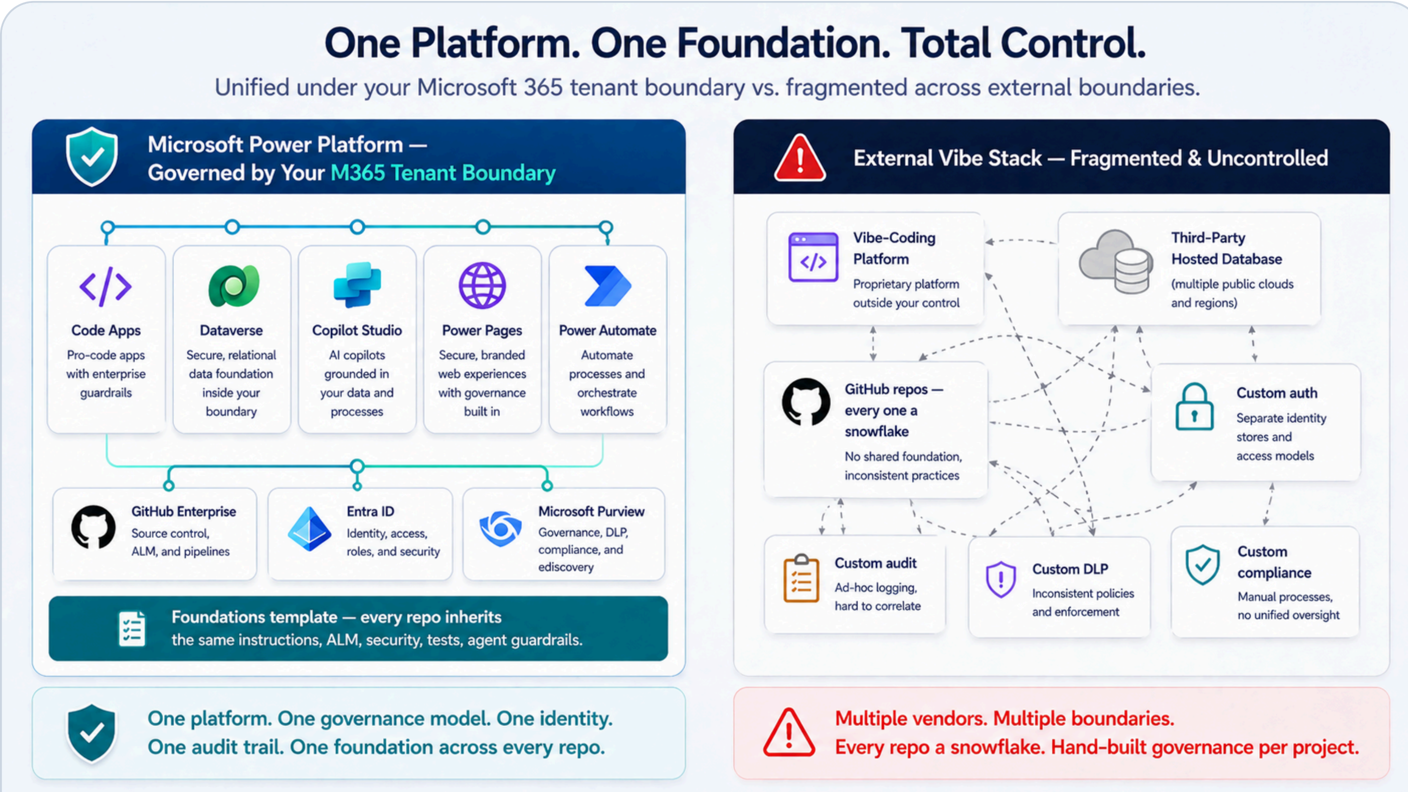
Code Apps — Open & Transparent

- ✓ **Named, versioned models** — you always know what wrote your code
- ✓ Any model in VS Code or GitHub Copilot works here
- ✓ Switch per repo, per developer, per task — even per file
- ✓ GitHub Enterprise controls keys, models, audit, policies
- ✓ `copilot-instructions.md` sets guardrails across every assistant
- ✓ New frontier model today → available within days

External Vibe Platforms — Opaque Models

- ✗ **No model transparency** — you don't know which model or version
- ✗ Cannot verify whether you're on the latest frontier capability
- ✗ The platform vendor decides the model roster — switching is not yours
- ✗ No version history, no audit trail of which model produced what
- ✗ Enterprise key management lives outside customer control
- ✗ In a landscape that shifts weekly, opacity is a liability

Fragmentation feels fast at first.
Platform coherence wins at scale.



Unified platform vs fragmented tools — connected, governed, reliable, built to scale.

CHAPTER 10 · FOUNDATIONS

Getting started is the hardest part — until now.

Most teams spend their first month discovering platform quirks the hard way. **PAppsCAFoundations** encodes every one of those lessons into a repeatable, tested path — so your team builds features on day one, not debugs platform quirks.

PAppsCAFoundations — a proven foundation, a clear path, faster to value.

🧭

Consulting-Grade Methodology

Business decomposition → scope refinement → Dataverse planning → prototype validation → build & deploy. Encoded in your repo.

🤖

Agent-Guided Development

14 scoped instruction files plus `AGENTS.md` — Copilot, Cursor, Claude Code, every agent follows the same rules.

⚡

Interactive Setup Wizard

A 9-step wizard verifies prerequisites, scaffolds React + Fluent UI + Dataverse, seeds prototype assets, runs smoke tests.

🧪

Testable from Day One

Vitest, Playwright, MSW ship in the scaffold. Smoke tests pass on first run. Mocks let you test before Dataverse exists.

🔐

Security-First Auth

1Password CLI integration or AES-256-GCM encrypted secrets. Pre-commit hooks block accidental secret commits.

🔄

Foundations Sync

Pull the latest instructions, scripts, and wizard into your downstream project. No fork management. No merge conflicts.

OPEN SOURCE · MIT LICENSED

Start strong. Build smart. Deliver impact.

Clone the template, run the wizard, start building. Your team gets the accumulated lessons of every team that came before.

Open Repo → (https://github.com/martycarreras-psnl/PAppsCAFoundations)

See Setup Path (https://martycarreras-psnl.github.io/PAppsCAFoundations/guide.html)

Start Building (https://aka.ms/codeappsfoundations)

CHAPTER 11 · THE VERDICT

If you're building serious enterprise applications, platforms win.

Build with AI speed. **Govern with enterprise confidence.**

Choose Code Apps when you value:

- Tenant-resident, BAA-covered data (PHI, PII, regulated)
- A maturity model from citizen to pro developer
- Native MCP and agent intelligence
- Configure-once governance for 1,000+ makers
- External-facing apps via Power Pages BYOC
- Freedom to use any frontier AI model in VS Code

Reconsider external vibe platforms when:

- PHI or other regulated data is in scope
- You need tenant-wide governance and ALM
- Repos that don't inherit a common foundation become unmanageable
- Agent integration is a strategic priority
- Long-term maintainability matters more than first-week velocity

"The AI coding tool is just the hammer. The platform is the house."

For organizations that value governance, compliance, agent integration, long-term maintainability, and the ability to scale citizen development safely — **especially when PHI is involved** — Power Apps Code Apps with Dataverse is the clear, defensible enterprise choice.

Start with Foundations →

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DISCLAIMER

This content is **not a formal statement, position, or endorsement from Microsoft**. It is an observational perspective from Marty Carreras (https://www.linkedin.com/in/carrema/), Microsoft's US Healthcare & Life Sciences AI Business Process and Workforce CTO, offered to help organizations think through enterprise application strategy. Opinions are the author's own.

Power Apps Code Apps Advantage · Build with AI speed. Govern with enterprise confidence.